

CO₂ Laser Cutter Procedures

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1. Introduction

This document outlines the procedures for making use of the CO₂ laser cutter. Firstly, some important decisions need to be made when considering the use of the laser cutter. Thereafter, the specifications of the available laser cutter and drawing and setup processes are discussed. Lastly, the procedure for using the laser cutter facilities is described.

2. Suitability for CO₂ laser cutting

There are many aspects to consider when designing a component, and several of these aspects directly influence the manufacturing method that will be required. Some points to consider before deciding on using the laser cutter:

- Can the component be purchased off the shelf?
- Can the component easily be manufactured through another available process?
- How many parts will be manufactured?
- Material choice (limited to 3 mm and 6mm supplied MDF for Mechatronics 424). Is the machine able to cut the chosen material? Common materials are acrylic, wood, MDF, rubber, PVC and paper.
- Can the cost of manufacture be justified by the intended function of the component?
- Can the part easily fit within the machine dimensions?
- Are the machine tolerances acceptable for the final product?

3. Laser specifications

This section covers some of the basic laser specifications as well as considerations for the design and manufacturing of the part.

3.1. Laser cutting process

A CNC laser cutter is a computer controlled cutting machine that forms the desired part by cutting or engraving on a base material (typically sheets) using focused light. The light is formed in a gas filled tube which is then reflected via mirrors to the machine gantry and directed downwards through a focussing lens. A typical CO₂ laser cutter consists of 2 to 3 axes. Although the department's machine has a z-axis, this axis is primarily used to set the initial height and focus before cutting or engraving commences. The machine can be controlled manually to some degree, however in most circumstances RDWorks V8 will be used to create the necessary toolpaths from a CAD model or from drawings made from within RDWorks. RDworks will create or send a file that the controller reads in order to move and control the machine to produce the desired part.

3.2. CO₂ laser cutting machine

The facility has one Advanced Machinery (AM) TruCUT 150W CO₂ laser cutter (Figure 1 and Figure 2). The machine specifications are summarised in Table 1, but may differ slightly from the values stated.



Figure 1: AM CO₂ laser cutter.

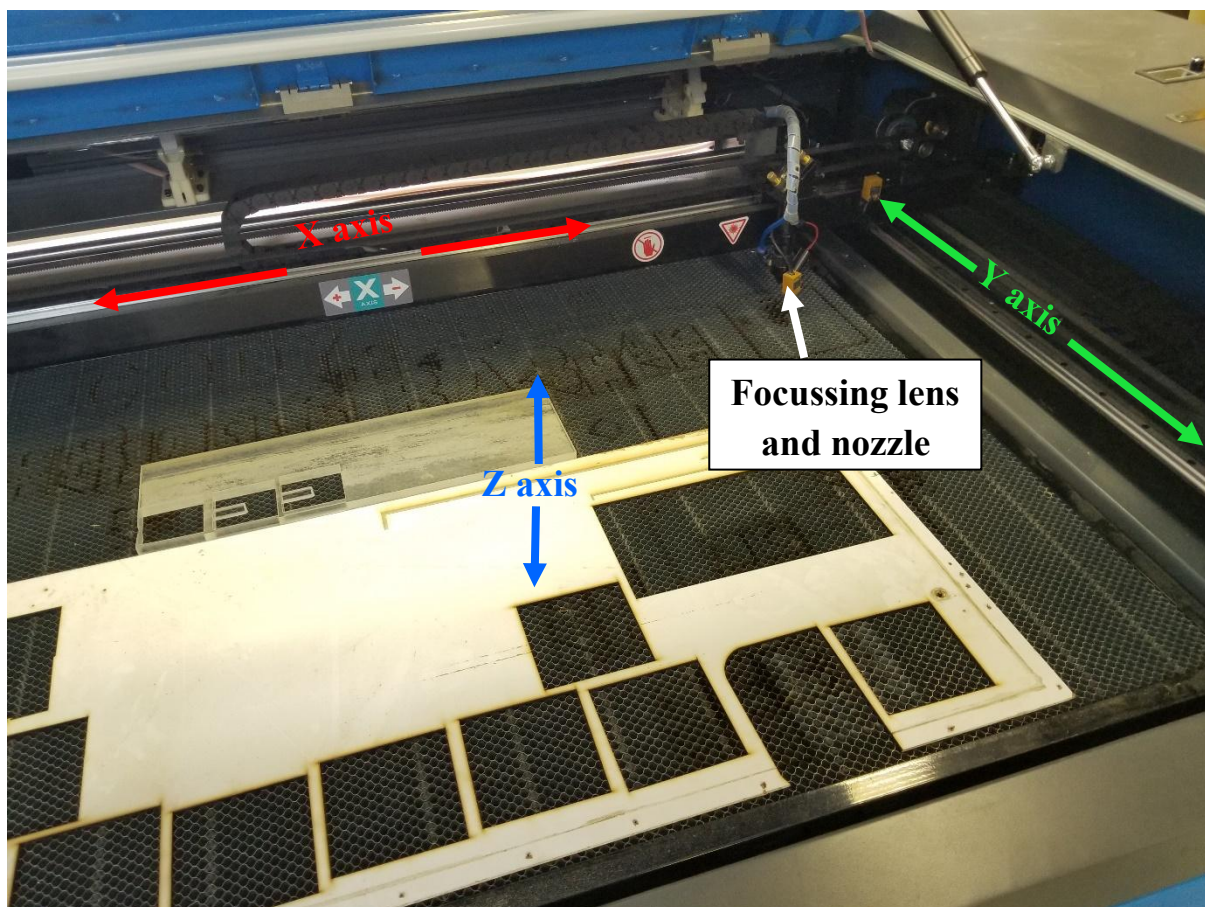


Figure 2: AM CO₂ laser cutter axes.

Table 1: AM CO₂ laser cutter specifications

Bed size	1300 mm by 900 mm (x, y)
Laser power	150 W, max usable ~ 100 W
Laser nozzle to bed clearance	< 320 mm (approximately)
Max travel speed	250 mm/s
Cutting speed	1 mm/s to 250 mm/s (material dependant)
Part accuracy	+/- 0.2 mm (approximately)
CNC machine controller	RUIDA controller
Program language	RDWorks V8, DXF file

3.3. Design and manufacturing considerations

Special care should be taken in the design of parts to make them suitable for CO₂ laser cutting and to ensure clarity and ease of set up for the artisan operating the machine. The most important design aspects are discussed in this section.

3.3.1. Material type

The laser cutter is capable of cutting a wide variety of materials, however, each material generally has different feedrates and cutting power settings which need to be used. Research into feasible materials and thicknesses should be conducted before ordering or attempting to cut on the machine. For example, it is possible to cut MDF wood on the machine, but the success of this depends on the density of the wood and whether there is significant moisture present in the stock.

3.3.2. Kerf

Kerf is the material lost due to the cutting process. It is important to take this into account especially if the designer is aiming to fit parts together or to use both parts on either side of the cut (Figure 3). The material lost is approximately 0.1 mm and this can be compensated for in software to be on centre or to either side of the profile depending on the designer's requirements.

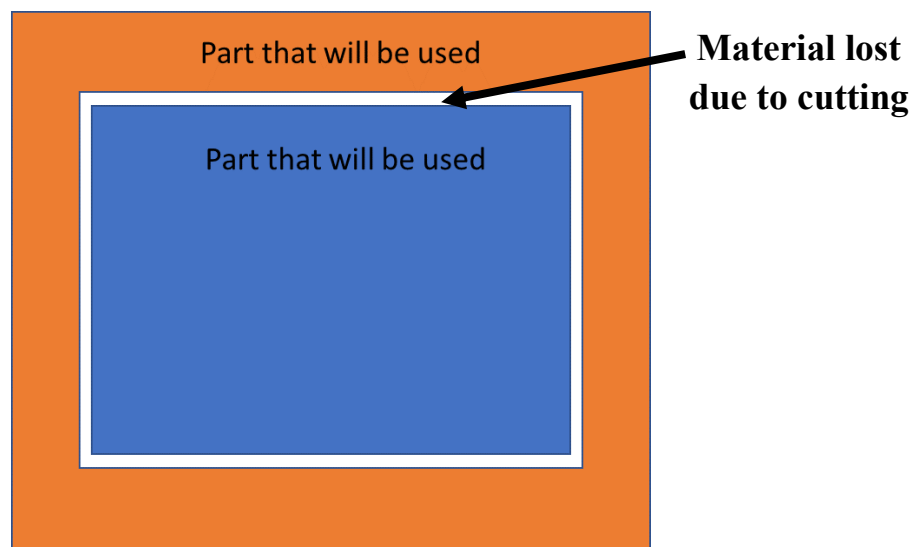


Figure 3: Example of kerf and how it can affect part sizes.

3.3.3. Bevel and accuracy

Depending on the material and thickness cut, a bevel on the cut edges will generally be observed. The designer should be aware of this and either account for post cutting operations to correct this or ensure it does not impact the final assembly and fit.

The accuracy of the machine is approximately ± 0.2 mm, however it should be possible to achieve ± 0.1 mm with a few cutting iterations.

3.3.4. Entry and exit points

In order to begin and end the cutting of a profile, the machine makes use of entry and exit points. The designer should take note that the points will be usually be outside of the part for outer perimeters and inside of any inner perimeters. This is important for selecting stock size and if the designer intended on using any parts nested within the model. The entry and exit points can be removed, allowing the cut to begin on the actual part line, however this usually affects the cut quality at those specific points. Figure 4 shows a part with multiple entry and exit points. These are indicated by the short red lines in RDWorks and have been circled in green for clarity.

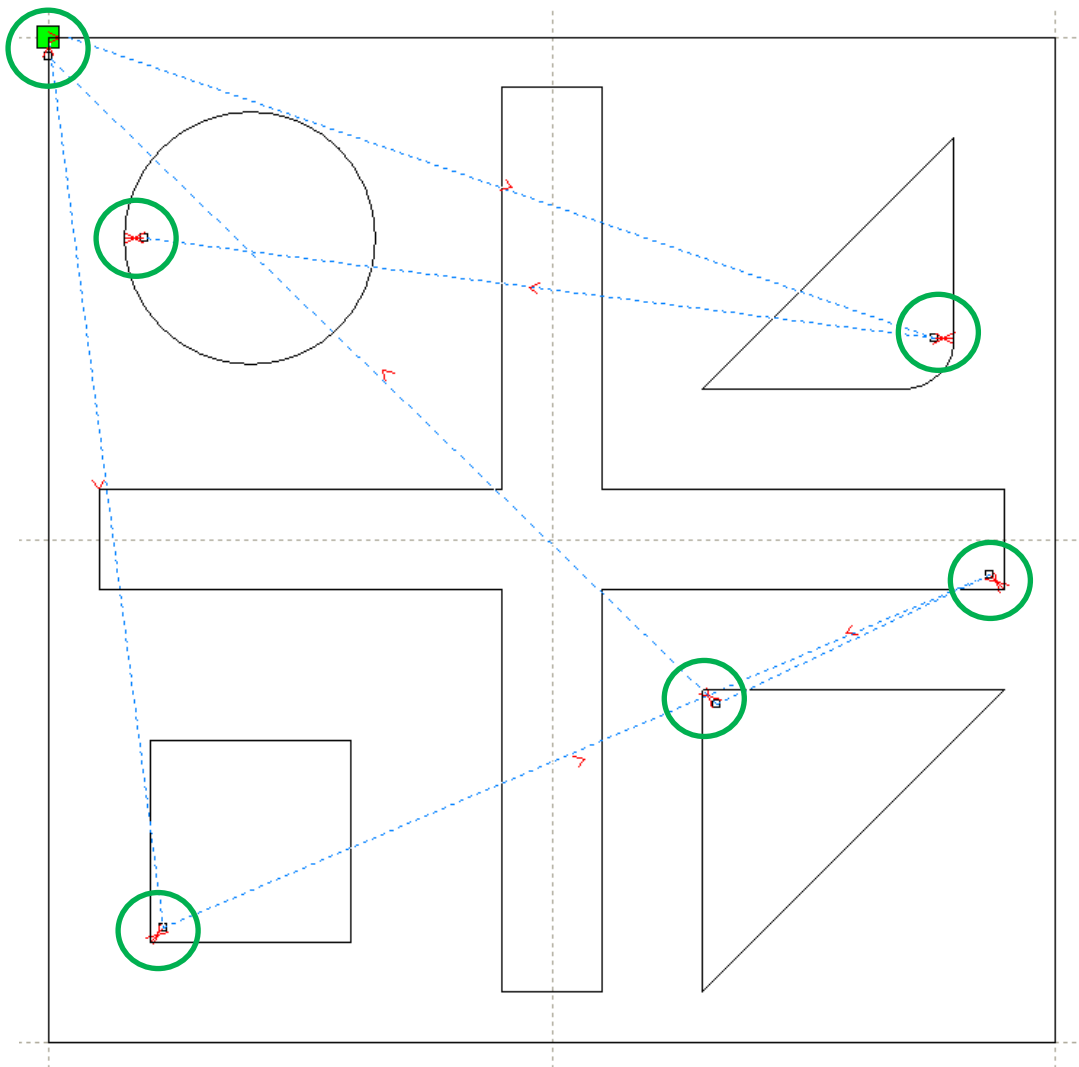


Figure 4: Sample part showing multiple entry and exit points (circled in green).

3.3.5. Support or tabs

Depending on the part geometry, tabs or support may be necessary to keep the part stable during machining and prevent the part from completely breaking loose (Figure 5). Generally, tabs can be avoided by ensuring that the cutting order is in such a way that if the part does break loose that it will only be after that area is completed. This is more often an issue on very small parts as they can either partially fall through the grid or honeycomb surface (possibly causing a collision with the nozzle) or completely fall through before the next operation, thereby going uncut. Tabs are usually removed manually once the cutting operations have been completed.

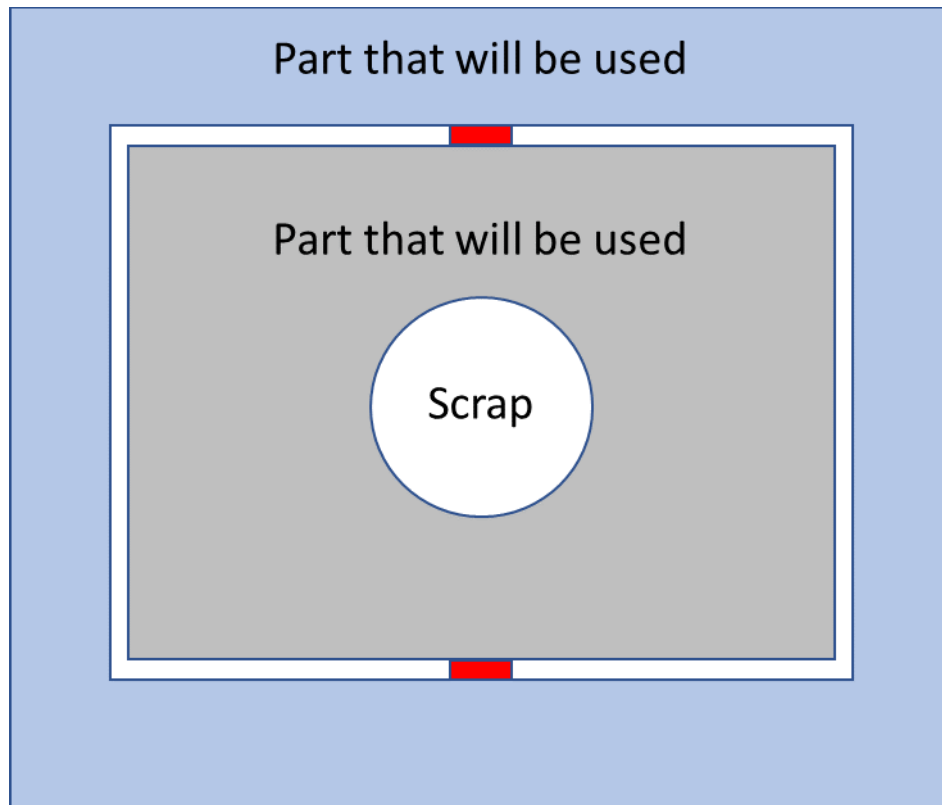


Figure 5: Example use of tabs or supports. Tabs are shown in red.

3.3.6. Stock size

Stock is raw material that the part will be manufactured from. Considerations when ordering stock are; maximum working area of the machine, allowance for entry and exit points (especially if multiple parts are cut from the same sheet), allowance for test cuts (should the cutting parameters not be known) and the necessity for tabs or supports.

3.3.7. Finishing/post processing

Depending on the designers desired finish, manual finishing of the part may still be required. These processes can vary from a light sanding to milling edges or drilling and tapping holes.

4. Procedure

The procedure that must be followed through the design, preparation and request stages, is outlined in this section. Should your cutting request fall outside of these specifications, please contact Xola Madyibi (xola@sun.ac.za) with your request. A meeting may be necessary to discuss whether the request is possible/feasible.

4.1. CAD preparation

Parts can be designed in Autodesk Inventor or your preferred CAD package. The RDWorks program requires a .DXF file. The DXF file should only contain the 2D view of your part with no annotations. Double check that the scaling is correct in the DXF file.

4.2. Download and install the latest version of RDWorks from the laboratory info

Sharepoint: [CO2 Laser Cutter](#)

4.3. Import and layout parts

Import the DXF files and layout your parts in RDWorks. Keep in mind the in-house wood sheet size is only 900 mm x 600 mm. Save separate “. rld” files if cutting both 3 mm and 6 mm sheets.

4.4. Contact technician

Contact Xola Madyibi (xola@sun.ac.za). Have the following prepared:

- Reason for using CO₂ laser cutter. Please perform background reading and research on the process.
- Proposed layout of parts for in “. rld” format.

4.5. Order stock

In-house stock may be used, however it is limited to 3 and 6 mm MDF (900 mm x 600 mm).

4.6. Part checking and scheduling

The technician will briefly check the layout. If satisfactory and approved, cutting will commence as soon as possible.

4.7. Costing

The costs are shown in Table 2. A quotation for cutting will be sent and must be approved by your supervisor before cutting may commence.

Table 2: Laser cutting costs

Admin fee (layout and setup)	R50 per job
Cutting cost	R6 per minute of cutting (3 mm cut at 20 mm/s and 6 mm cut at 10 mm/s)
Material 3 mm x 900 mm x 600 mm	R50
Material 6 mm x 900 mm x 600 mm	R90

4.8. Part collection

Once the parts are complete you will be contacted via email to arrange a collection date and time.